Building Science

Lecture 9: Artificial Lighting

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Lecture outline

- Colour
- Effects of different artificial lighting types
- Lumen Method

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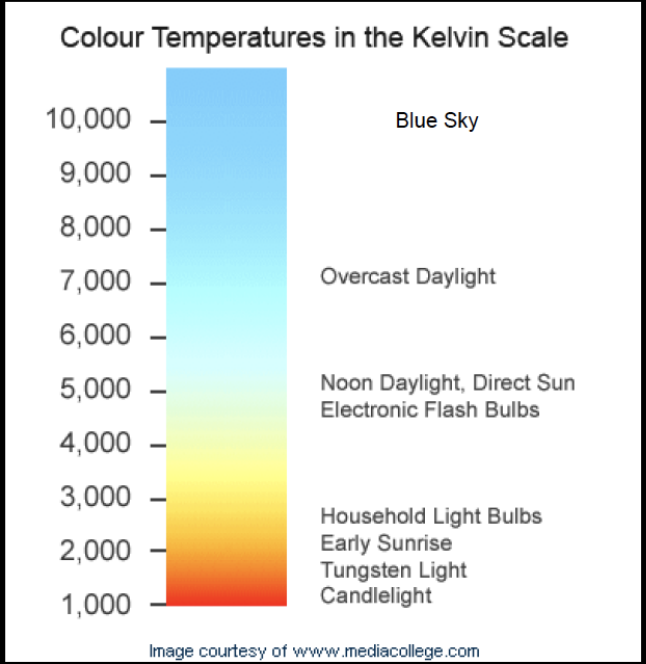
Colour: colour appearance

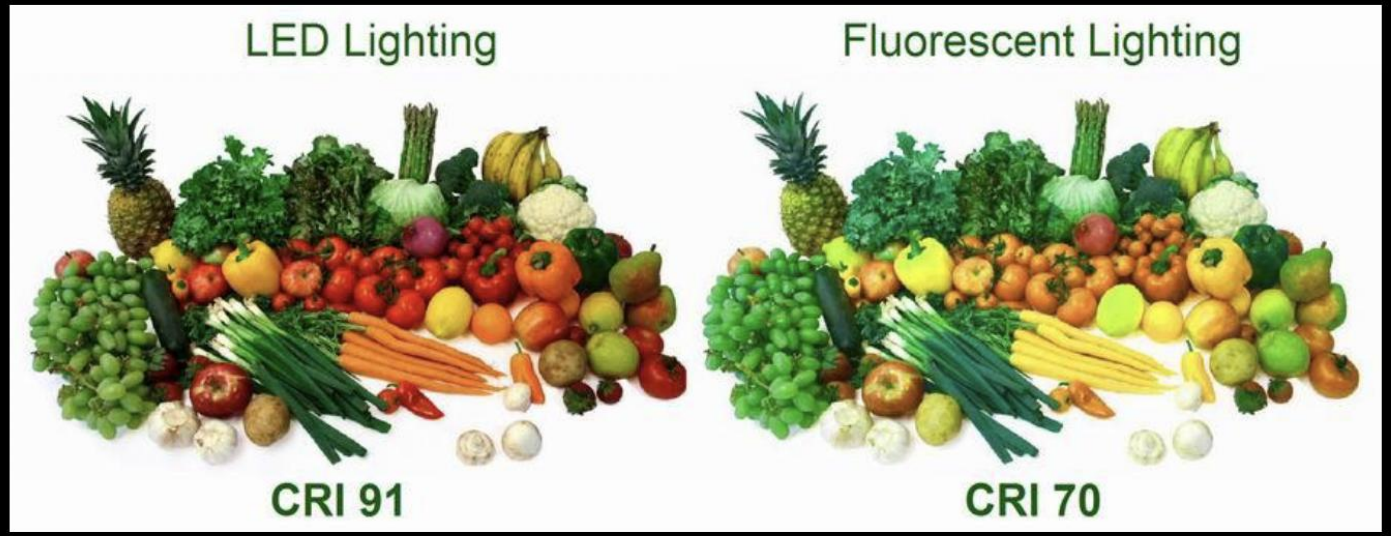
- ✓ classified according to the correlated colour temperature scales (CCS)
- ✓ measured in Kelvin (K)
- ✓ three main classifications:

Classification	*K*
Warm	3300 K
Intermediate	3300 - 5300 K
Cold	5300 K

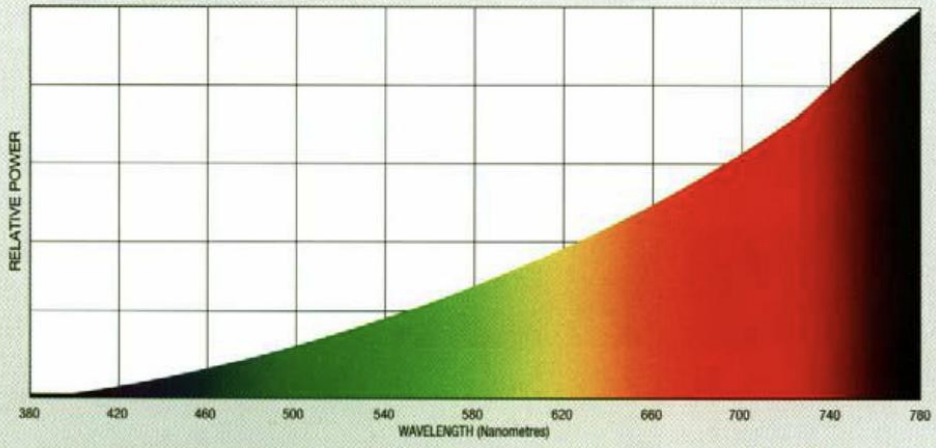


Colour: colour rendering

- ✓ quantified by the colour rendering index (CRI)
- ✓ compares surfaces under a particular light to their appearance under a reference light (an ideal or natural light)
- ✓ good rendering means a high CRI, with a maximum of 100 indicating that the source renders all colours identical to the reference

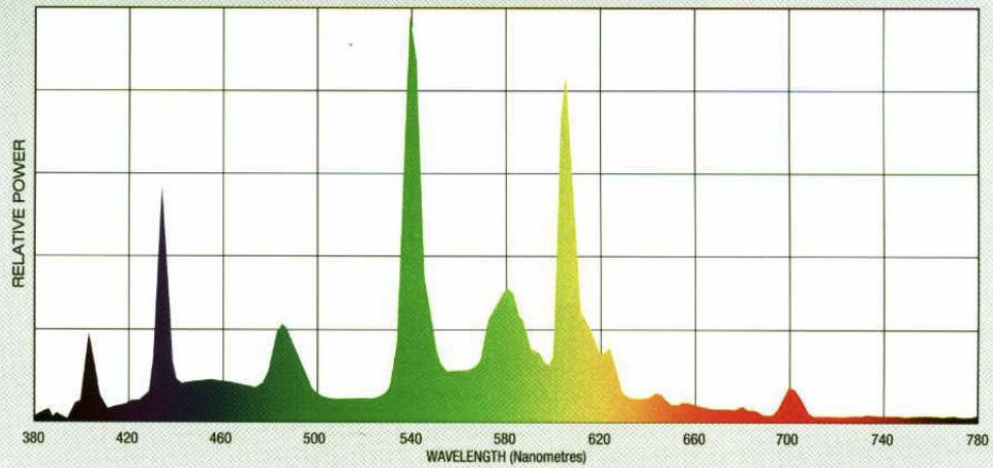


Colour: Incandescent lamp spectrum



Incandescent light is produced by heating a solid object—the filament—to the glowing point. In a sense, this is the way light is produced by the sun. It follows, therefore, that both standard incandescent and halogen lamps exhibit smooth, even SPD curves, not unlike sunlight.

Colour: Fluorescent lamp spectrum



The spectral energy from a fluorescent lamp is produced by two separate mechanisms: the mercury arc generates narrow bands of energy—which appear as spikes—and the fluorescence of the phosphor coating produces a more balanced, continuous spectrum of visible light. Because of its special triphosphor coating, the energy bands of this OCTRON® 4100 lamp appear at the specific frequencies of red, green and blue, the primary colours of light.

Lamp

✓ the artificial **light source**

Electrical Light Sources for Buildings

- **INCANDESCENT**
  - INCANDESCENT
  - HALOGEN
- **DISCHARGE**
  - FLUORESCENT
    - *COMPACT FLUORESCENT*
  - HIGH INTENSITY DISCHARGE
    - *MERCURY*
    - *METAL HALIDE*
    - *HIGH PRESSURE SODIUM*
- **SOLID STATE**
  - LED

Source: Lechner (2009)

Lamps

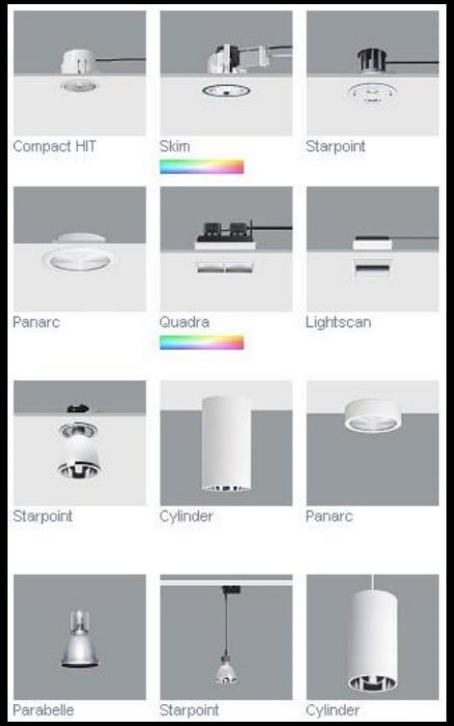
Lamps: Comparison of the major lamp groups

Source: Lechner (2009)

Lamp group	Efficacy (Lumens/Watt)	Life (Hours)	applications	characteristics
Incandescent	8-20	800-3,000	Decorative light	Obsolete
Halogen	12-24	2,000-5,000	Spotlighting	Excellent colour rendition, low initial cost
Fluorescent	30-105	10,000-30,000	Diffused lighting of a large are (offices, schools, residential, industrial)	Sensitive to temperature, not used outdoors in cold climate
Compact fluorescent	40-90	8,000-10,000	Replaces incandescent and halogen lamps	Relatively small size
Metal Halide	50-115	10,000-25,000	Diffused lighting or wide beams (offices, stores, schools, industrial, outdoor)	5 to 10 minute delay in start or restart Contain mercury
High Pressure Sodium	80-140	24,000-30,000	Diffused lighting or wide beams where colour is not important (industrial, outdoor, warehouses)	Poor colour rendition About 5 minute delay in start or restart
LED	10-60	10,000-50,000	Decorative, accent lighting Exit, emergency and wayfinding	Excellent source of pure colour (red, green, blue etc)

Luminaires

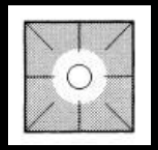
- ✓ includes reflectors, diffusers, mechanical fixtures and electrical connections



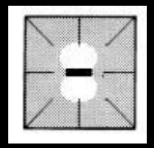
Luminaires: Lighting fixtures

✓ Direct

✓ Semi-direct

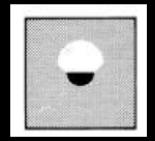


✓ General diffuse



✓ Direct-indirect

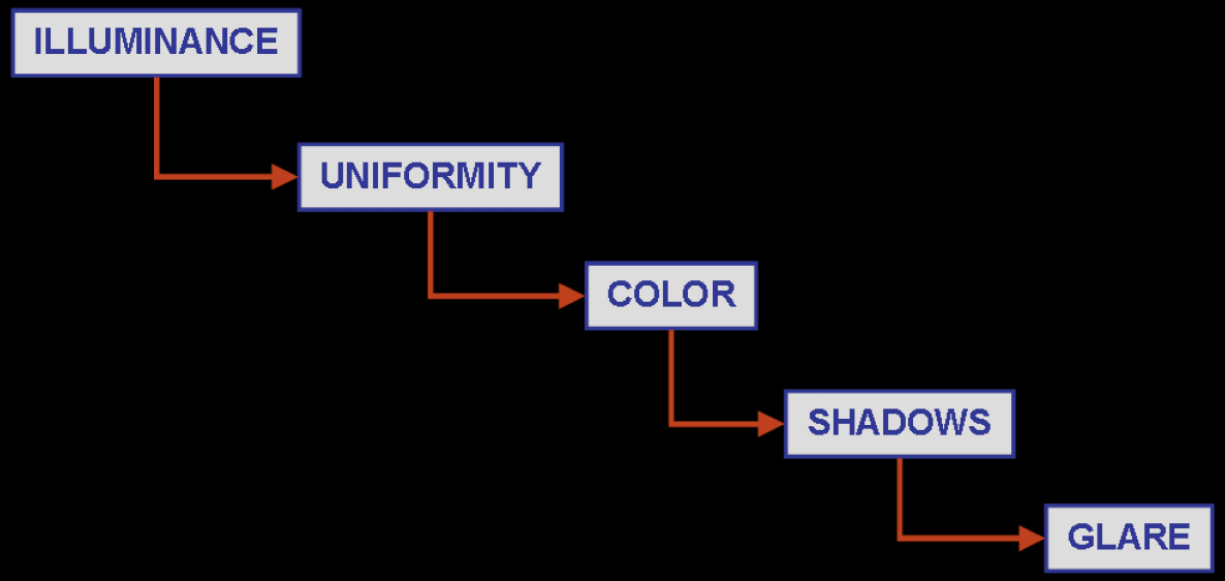
✓ Semi-indirect



✓ Indirect

Lighting Design: Quantitative approach

Performance Approach



Lighting Design: Quantitative approach

TASK REQUIREMENT	EXAMPLE	GUIDELINE ILLUMINANCE
General lighting Simple visual tasks	Access routes	50
	Staircases, corridors	100
	Bathrooms	150
	Foyers, lobbies	200
General Lighting Working spaces	Classrooms	300
	General Offices	500
	Drawing offices, studios	700
	Electronic Assembly	1000
Lighting for complicated visual tasks (small detail and low contrast)	Fine Work and Inspection	1500
	Assembly of minute mechanisms	2000

Lighting Design: Quantitative approach

Uniformity

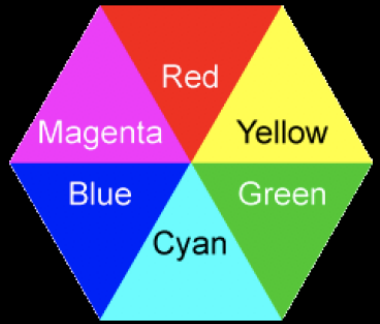
✓Overall lighting too bright for other activities?

✓Effect on spatial character?

Lighting Design: Quantitative approach

Colour

Hue, Chroma, Value



Lighting Design: Quantitative approach

Shadows

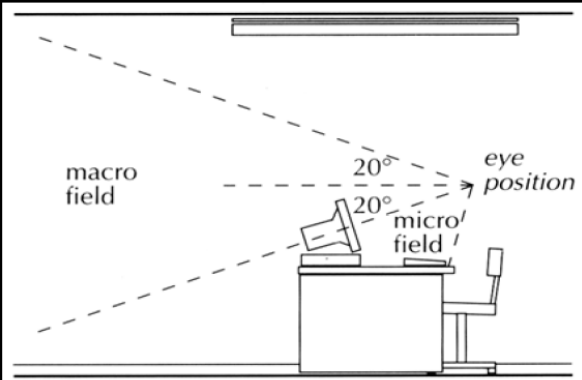
✓Texture, character, visual interest

✓Task performance, additional task lighting

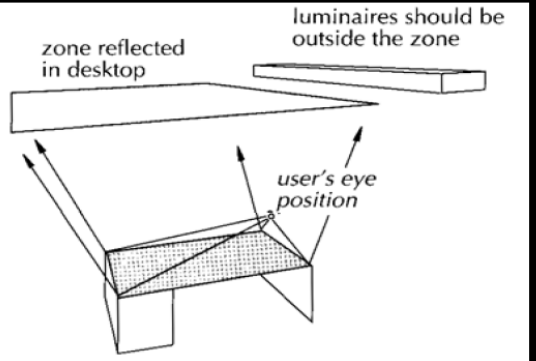
Glare

✓Task performance, repositioning lamps

✓Direct glare and veiling reflections



Macro and micro fields of view.



Zone where images of luminaires are reflected in glossy desktop surfaces.

Lighting Design: Qualitative approach

Architectural Lighting

- ✓ **Ambient Lighting**
  - Defines spatial appearance / character
  - Forming spatial zones and borders
  - Establish continuity
- ✓ **Accent Lighting**
  - Emphasis on element
  - Establish hierarchy
- ✓ **Task Lighting**
  - Satisfy functional requirements
  - Orientation and safety
- ✓ **Decorative Lighting**
  - Narrative and effects

Lumen Method

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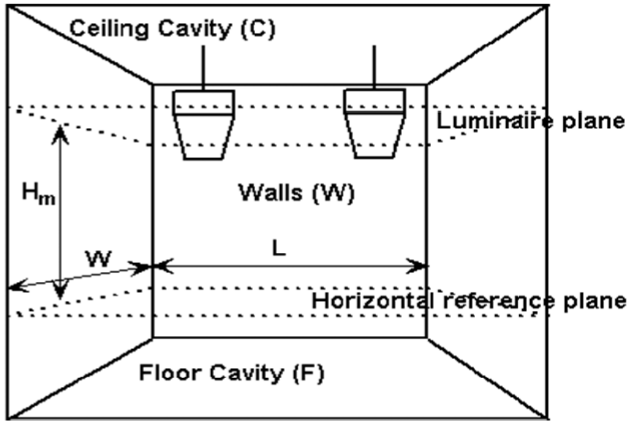
Lumen method

- ✓ applicable to a large number of real-life situations, e.g. offices and classrooms

- ✓ used to determine the **number of lamps** should be installed for a given area or room

- ✓ **a uniform distribution** of a large number of fixtures i.e. at the point of least illumination it is not less than 70% of the maximum

  » ***spacing***



Lumen method: Basic equation

Since illuminance is a measure of the luminous flux or rate of light flow falling on a surface area, we can write the basic expression

"illuminance equals luminous flux per surface area."

Illuminance = Luminous Flux / Area

$$\textbf{Lux} = \textbf{Lumens} / m^2$$

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Lumen method: Lumen equation

In practical terms, we aim for a certain lux level for the workplane, say 300 lux for an office. We determine the area to be lighted.

By knowing the two necessary elements,

we can rewrite our equation.

Luminous Flux = Illuminance x Area

$$\textbf{Lumens} = \textbf{Lux} \times \textbf{m}^2$$

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Lumen method: Lumen equation

In this example,
let's use an office of 5m x 4m.

We would need

300 x 20 = 6000 lumens
Lux x m^2 = Lumens

Now just pick the number and type of fixtures that will provide enough lumens

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The Lumen Method formula:

$$N = \frac{E \times A}{F \times UF \times MF}$$

Where,

N = number of lamps required

E = illuminance level required (lux)

A = area at working plane height (m^2)

F= average luminous flux from each lamp (lm)

UF=utilisation factor, an allowance for the light distribution of the luminaire and the room surfaces

MF=maintenance factor, an allowance for reduced light output because of deterioration and dirt

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Thank you

Next time: Acoustics and noise control

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